

$$\begin{bmatrix} 1 \\ 0 \\ 2 \\ -3 \end{bmatrix}$$

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→ Mat
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$\therefore$ a system of linear equation is said to be consistent if it posses a solution otherwise it is inconsistent.

→ Matrix form of system of linear equations.

In the above equation ① let $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

∴ equation ① becomes $AX = B$

Matrix A is called the coefficient matrix of the given system, the matrix X is called the matrix of unknowns or matrix of the variables and matrix B is called the matrix of independent terms or constants.

Note:- consider a linear homogeneous equation

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= 0 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= 0 \end{aligned} \right\} \text{②}$$

Are of m homogeneous linear equation in n unknowns.

∴ the above system can be written as $AX = 0$ where 0 = zero matrix of order $m \times n$. In the above system is always consistent as $x_1 = 0, x_2 = 0, \dots, x_n = 0$ is always a solution of equation ② therefore this solutions are called trivial solutions, any other solution is called non-trivial solution.

working rule of finding the solution of homogeneous system of equation i.e. $AX=0$

consider the matrix form of homogeneous system of equation. $AX=0$

(i) to reduce the coefficient matrix.
(ii) to a row reduced echelon matrix — matrix ~~(iii)~~ by applying the elementary row operation and hence write the equivalent system of equation.

(iii) If rank of a matrix $A = r$ the given system is reduced to r equations

(iv) if $r = n$ the only solution will be trivial solution i.e. $x_1 = x_2 = x_3 = \dots = x_n = 0$

(v) if $r \neq n$ then the solution will be dependent on $n - r$ parameters in terms of which a complete solution can be obtained.

→ Solve for the system of equation.

Ex:- $3x_1 = 0$

$$-x + 4x_2 = 0$$

$$4x_1 - 4x_2 + 3x_3 = 0$$

⇒

$$AX=0$$

$$\begin{bmatrix} 3 & 0 & 0 \\ -1 & 4 & 0 \\ 4 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_1 \rightarrow R_1 \times 1/3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 4 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$AX=0$$

system of

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$$x_n=0$$

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ed.

$$R_2 \rightarrow R_2 + R_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 4 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 \times 1/4$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_3 \rightarrow R_3 + 4R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_3 \rightarrow R_3 \times 1/3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$\therefore$ Equivalent eqn's are :-

$$x_1 + 0x_2 + 0x_3 = 0$$

$$\Rightarrow x_1 = 0$$

$$0x_1 + 1x_2 + 0x_3 = 0$$

$$\Rightarrow x_2 = 0$$

$$0x_1 + 0x_2 + 3x_3 = 0$$

$$\Rightarrow 3x_3 = 0$$

$$\rho(A) = \mu = 3$$

Rank of matrix

$n = 3 \rightarrow$ no. of unknown variables

$$\therefore n = \mu$$

$$\therefore x_1 = 0, x_2 = 0, x_3 = 0$$

$\therefore$ it is trivial solⁿ.

$$Q2:- x + y - 6z = 0$$

$$-3x + y + 2z = 0$$

$$x - y + 2z = 0$$

$$\begin{bmatrix} 1 & 1 & -6 \\ -3 & 1 & 2 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 + 3R_1$$

$$\begin{bmatrix} 1 & 1 & -6 \\ 0 & 4 & -16 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 1 & -6 \\ 0 & 4 & -16 \\ 0 & -2 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 + 2R_3$$

$$\begin{bmatrix} 1 & 1 & -6 \\ 0 & 0 & 0 \\ 0 & -2 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & -6 \\ 0 & -2 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 \times -1/2$$

$$\begin{bmatrix} 1 & 1 & -6 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

= 0

The equivalent eqns are -

$$1x + 1y - 6z = 0 \rightarrow (1)$$

$$0x + y - 4z = 0 \rightarrow (2)$$

$$\therefore \rho(A) = n = 2$$

$$n = 3$$

$$\therefore n \neq \rho$$

$$\Rightarrow n - \rho = 3 - 2 = 1$$

$$\text{let } z = t$$

$$(2) \Rightarrow y - 4z = 0$$

$$y - 4t = 0$$

$$y = 4t \rightarrow (3)$$

by (3)

$$(1) \Rightarrow x + y - 6t = 0$$

$$x + 4t - 6t = 0$$

$$\boxed{x = 2t}$$

$$x = 6t - 4t$$

$$\therefore x = 2t, y = 4t, z = t$$

$\therefore$ it is non trivial.

$$\begin{aligned} \text{Q:- } & 2x_1 + x_2 + 3x_3 = 0 \\ & x_1 + 2x_2 = 0 \\ & x_2 + x_3 = 0 \end{aligned}$$

$$\Rightarrow AX = 0$$

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$(i) R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$(ii) R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$(iii) R_2 \rightarrow R_2 \times (1/3)$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$(iv) R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\rho(A) = r = 3$$

$$n = 3$$

$\therefore$ If $n = r$

The equivalent eqns are :-

$$x_1 + 2x_2 = 0 \quad \text{--- (1)} \rightarrow x_1 = 0$$

$$x_2 - x_3 = 0 \quad \text{--- (2)} \rightarrow x_2 = 0$$

$$2x_3 = 0 \quad \text{--- (3)}$$

$$\Rightarrow x_3 = 0$$

$\therefore$ It is a trivial ~~eqn~~ solⁿ.

$$Q:- x - 3y + 2z = 0$$

$$7x - 21y + 14z = 0$$

$$-3x + 9y - 6z = 0$$

$$\Rightarrow AX = 0$$

$$\begin{bmatrix} 1 & -3 & 2 \\ 7 & -21 & 14 \\ -3 & 9 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$(i) R_2 \rightarrow R_2 \times (1/7)$$

$$\begin{bmatrix} 1 & -3 & 2 \\ 1 & -3 & 2 \\ -3 & 9 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$(ii) R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & -3 & 2 \\ 0 & 0 & 0 \\ -3 & 9 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$(iii) R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & -3 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\therefore \rho(A) = n = 1$$

$$n = 3$$

$$\therefore n \neq \rho, n - \rho, 3 - 1 = 2$$

$\therefore$ The equivalence eqn is -

$$x - 3y + 2z = 0 \quad \text{--- (1)}$$

let us assume $z = t_1$ & $y = t_2$

$$\Rightarrow \textcircled{1} \rightarrow x - 3t_2 + 2t_1 = 0$$

$$\Rightarrow x = 3t_2 - 2t_1$$

$$\Phi: -x_1 + x_2 - x_3 + x_4 = 0$$

$$x_1 - x_2 + 2x_3 - x_4 = 0$$

$$3x_1 - x_2 + x_4 = 0$$

$$\Rightarrow AX = 0$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$(i) R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 3 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$(iii) R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & -4 & 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$(iv) R_2 \rightarrow R_2 \times (-1/2)$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & -3/2 & 1 \\ 0 & -4 & 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$(v) R_3 \rightarrow R_3 + 4R_2$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & -3/2 & 1 \\ 0 & 0 & -3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$\therefore \rho(A) = 3 = n$$

$$n = 4$$

$$\therefore n \neq n, n - n, 4 - 3 = 1$$

$\therefore$ The equivalence eqn is -

$$x_1 + x_2 - x_3 + x_4 = 0 \quad \text{--- (1)}$$

$$x_2 - 3/2 x_3 + x_4 = 0 \quad \text{--- (2)}$$

$$-3x_3 + 2x_4 = 0 \quad \text{--- (3)}$$

$$\textcircled{3} \Rightarrow -3x_3 + 2t = 0$$

$$\Rightarrow +3x_3 = +2t$$

$$\Rightarrow x_3 = \frac{2}{3}t$$

$$\begin{aligned} \Phi: -x_1 + x_2 - x_3 + x_4 &= 0 \\ x_1 - x_2 + 2x_3 - x_4 &= 0 \\ 3x_1 + x_2 + x_4 &= 0 \end{aligned}$$

$$\Rightarrow AX = 0$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$(i) R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 3 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$(ii) R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & -2 & 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$\therefore \rho(A) = 2$$

$$n = 4$$

$$\therefore n \neq h, n - h = 4 - 2 = 2$$

Equivalent eqn are -

$$x_1 + x_2 - x_3 + x_4 = 0 \quad \text{--- (1)}$$

$$-2x_2 + 3x_3 - 2x_4 = 0 \quad \text{--- (2)}$$

Let us assume $x_3 = t_1$ & $x_4 = t_2$

$$(2) \Rightarrow 2x_2 + 3t_1 - 2t_2 = 0$$

$$\Rightarrow 2x_2 = 2t_2 - 3t_1$$

$$\Rightarrow x_2 = \frac{2t_2 - 3t_1}{2}$$

$$\textcircled{1} \Rightarrow x_1 + \left(\frac{2t_2}{2} - \frac{3t_1}{2} \right) - t_1 + t_2 = 0$$

→ System of non-homogeneous linear equations

Consider, the system of m non-homogeneous equation in n unknowns i.e. $AX=B$ where, A is the coefficient matrix, X is the matrix of unknown, B is matrix of independent terms.

$$(A:B) = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} : b_1 \\ a_{21} & a_{22} & \dots & a_{2n} : b_2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} : b_m \end{bmatrix} \therefore \text{is known as a augmented matrix.}$$

→ working rule to find the solution of system of non-homogeneous equation i.e. $AX=B$

- (i) consider the system of equation $AX=B$.
- (ii) find the rank of the augmented matrix ~~$AX=B$~~ $(A:B)$ by echelon form.
- (iii) then, from the above step rank of A can be obtained.
- (iv) If rank of $A < \rho(A:B)$ then the system is inconsistent and it has no solution.
- (v) If $\rho A = \rho(A:B)$ then the system is consistent then we can find the solution by considering the equivalent system of equation.

$$\begin{aligned} \text{Q- } x + y - 2z &= 5 \\ x - 2y + z &= -2 \\ -2x + y + z &= 4 \end{aligned}$$

consider $AX = B$

$$(A:B) = \left[\begin{array}{ccc|c} 1 & 1 & -2 & 5 \\ 1 & -2 & 1 & -2 \\ -2 & 1 & 1 & 4 \end{array} \right]$$

(i) $R_2 \rightarrow R_2 - R_1$

$$(A:B) = \left[\begin{array}{ccc|c} 1 & 1 & -2 & 5 \\ 0 & -3 & 3 & -7 \\ -2 & 1 & 1 & 4 \end{array} \right]$$

(ii) $R_3 \rightarrow R_3 + 2R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & -2 & 5 \\ 0 & -3 & 3 & -7 \\ 0 & 3 & -3 & 14 \end{array} \right]$$

(iii) $R_2 \rightarrow R_2 \times -1/3$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & -2 & 5 \\ 0 & 1 & -1 & 7/3 \\ 0 & 3 & -3 & 14 \end{array} \right]$$

~~related to first is related to second (dependent)~~

~~Moreover, I have 23 years of teaching experience.~~

$$R_3 \rightarrow R_3 - 3R_2$$

$$\sim \begin{bmatrix} 1 & 1 & -2 & : & 5 \\ 0 & 1 & -1 & : & 7/3 \\ 0 & 0 & 0 & : & 7 \end{bmatrix}$$

$$\therefore \rho(A:B) = 3$$

$$\rho(A) = 2$$

$$\therefore \rho(A) < \rho(A:B)$$

$$2 < 3$$

$\therefore$ It is inconsistent and has no solⁿ.

$$\begin{aligned} \textcircled{1} - \quad & x + y + z = -3 \\ & 3x + y - 2z = -2 \\ & 2x + 4y + 7z = 7 \end{aligned}$$

$$\Rightarrow AX = B$$

$$(A:B) = \sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 3 & 1 & -2 & : & -2 \\ 2 & 4 & 7 & : & 7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & -5 \\ 2 & 4 & 7 & : & 7 \end{bmatrix}$$

~~23(1)~~
~~23~~

$$R_3 \rightarrow R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & -5 \\ 0 & 2 & 5 & : & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 \times -1/2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & 1 & 5/2 & : & 5/2 \\ 0 & 2 & 5 & : & 5 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & 1 & 5/2 & : & 5/2 \\ 0 & 0 & 0 & : & 20 \end{bmatrix}$$

$$\rho(A:B) = 3$$

$$\rho(A) = 2$$

$$\therefore 2 < 3$$

$\therefore$ It is inconsistent and has no solnⁿ.

$$Q:- x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

$$\Rightarrow AX = B$$

$$\sim (A:B) = \begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 2 & 4 & -3 & : & 1 \\ 3 & 6 & -5 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - IR_1$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 2 & -7 & : & -17 \\ 3 & 6 & -5 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 2 & -7 & : & -17 \\ 0 & 3 & -11 & : & -27 \end{bmatrix}$$

$$R_2 \rightarrow R_2 \times (1/2)$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & -7/2 & : & -17/2 \\ 0 & 3 & -11 & : & -27 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & -7/2 & : & -17/2 \\ 0 & 0 & -1/2 & : & -3/2 \end{bmatrix}$$

$$\therefore \rho(A:B) = 3$$

$$\rho(A) = 3$$

$$\therefore \text{Id } \rho(A) = \rho(A:B)$$

$\therefore$ It is consistent and has solⁿ.

Equivalence eqns -

$$x + y + 2z = 9 \quad \text{--- (1)}$$

$$y + \frac{-7}{2}z = -\frac{17}{2} \quad \text{--- (2)}$$

$$-\frac{1}{2}z = -\frac{3}{2} \quad \text{--- (3)}$$

$$\textcircled{3} \Rightarrow z = 3$$

$$\textcircled{2} \Rightarrow y - \frac{7}{2}(3) = -\frac{17}{2}$$

$$\Rightarrow y = \frac{-17}{2} + \frac{21}{2}$$

$$= \frac{4}{2} \quad \# \quad \therefore y = 2$$

$$\begin{aligned} \textcircled{1} \Rightarrow x + 2 + 2(3) &= 9 \\ x &= 9 - 2 - 6 \\ &= 9 - 8 = 1 \end{aligned}$$

$$\therefore x = 1, y = 2, z = 3.$$